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FLIPPED CLASS ACTIVITY

Course Code & Title: CSE3006 Computer Networks

Name of the Faculty member: Dr.S.Periyanayagi, Senior Associate Professor/ SCAI

Class ID/ Slot: BL2026270100338/ B11+B12+B13

Class Strength / Number of Groups: 106 students / 21 Groups

Topic chosen: Recent Trends in Network Security

Justification for Choosing the Topic using Flipped Class:

The topic was chosen for flipped class is there are many recent trends in network security. The Unit 5 deals with the principles of networking applications and recent trends in network security. I taught of giving the topic 'Recent trends in network security' for flipped class.

This topic can be learnt in self-learning mode. As there are many methods of network security is used in various application. If this flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related to specific application. Thus 10 topics will be covered in 3 periods through this activity which is not possible in traditional teaching. This is the reason for choosing this topic for flipped class activity.

Goal (Learning Objective):

Objective: To explain the recent trends in network security for various fields.

Outcome: At the end of the flipped class activity

- The students will be able to explain the various trends in network security.
- The self-learning skill and team work skills will be enhanced.

Guideline for preparing the Activity:

1. Prepare a Power point presentation with minimum 20 slides.
2. Contents of Presentation:
    a. Title Slides – Topic, Team Members Name, Register Number, Class and Slot.
    b. 15 - 17 Slides – Content of the presentation
    c. Reference Slide
3. Submit the Power point presentation in MS teams once you have completed.

4. A group of 5 members per team should do the presentation.

Topic of Presentation:

Group 1: Recent Trends in Network Security Introduction & Cryptography
Group 2: Recent Trends in Network Security Introduction & Cryptography
Group 3: Substitution and Transposition Ciphers
Group 4: Substitution and Transposition Ciphers
Group 5: Data encryption standard (DES)
Group 6: Advanced encryption standard (AES)
Group 7: Data encryption standard (DES)
Group 8: Advanced encryption standard (AES)
Group 9: Breaking Data encryption standard
Group 10: Message Integrity - Cryptographic Hash Functions, Message Authentication Code
Group 11: Digital Signatures
Group 12: Public Key Cryptography - RSA Algorithm
Group 13: Public Key Cryptography - Diffie-Hellman algorithm
Group 14: Network layer security (IPSec)- IPsec and Virtual Private Networks (VPNs)
Group 15: Network layer security (IPSec)- The AH and ESP Protocols, Security Associations
Group 16: Network layer security (IPSec) - IPsec Datagram
Group 17: Network layer security (IPSec) IKE: Key Management in IPsec
Group 18: Network layer security (IPSec)- The AH and ESP Protocols, Security Associations
Group 19: Network layer security (IPSec)- IPsec and Virtual Private Networks (VPNs)
Group 20: Data encryption standard (DES)
Group 21: Advanced encryption standard (AES)

Plan for implementing this activity:

Outside class activity:

- Reading of learning material and Preparation of presentation: 10 days' time duration
- Discussions and clarification of doubts: Any time within the 1 weeks' time for discussion within the group mates and any clarification regarding the flipped class

Inside class activity:

- Presenting the topic assigned by each group – 15 minutes duration
- Question & answering, oral feedback – 3 minutes duration

Date of Submission and Evaluation in MS Team:

***CO-PO* Matrices:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
5	3	3	-	-	-	-	-	2	2	3	-	-

Evaluation of Flipped Classroom

Course Code & Title: CSE3006 Computer Networks
Name of the Faculty member: Dr.S.Periyanayagi
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Group	Register Number	Name of the Student	Topic	Content of Presentation (20)	Q & A (5)	Contribution (5)	Total (30)